

**ModulCode:-CMS3416-LOJ-S2-2526: Semantic Web**

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## Table of Contents

|                                                         |    |
|---------------------------------------------------------|----|
| 1. Introduction .....                                   | 3  |
| 1.1 Semantic web overview .....                         | 3  |
| 1.2 Semantic web technologies: RDF, OWL and SPARQL..... | 3  |
| 1.3 Role of Ontologies in Healthcare.....               | 3  |
| 2. Domain Description.....                              | 4  |
| 2.1 Healthcare information systems.....                 | 4  |
| 2.2 Key healthcare entities .....                       | 4  |
| 2.3 Relationships between healthcare entities .....     | 4  |
| 3. Domain Model .....                                   | 4  |
| 4. SPARQL queries .....                                 | 17 |
| 5. Conclusion .....                                     | 22 |
| References .....                                        | 24 |
| Appendix.....                                           | 26 |

## **1. Introduction**

### **1.1 Semantic web overview**

Semantic Web is an extension of the traditional World Wide Web, the role of which is to ensure accessibility to information in the internet to human beings, as well as to machines. In the traditional web system, the information is normally given in its human readable format. Such information is simple to read and understand by human beings, and a computer may literally fail to interpret a meaning behind it, especially in reading the information. Semantic Web addresses this weakness since it is used to present the information in structured formats whereby they stipulate a relation among data items .

### **1.2 Semantic web technologies: RDF, OWL and SPARQL**

Important in the achievement of this goal are the semantic Web technologies of RDF ( Resource Description Framework ), OWL ( Web Ontology Language ) and SPARQL ( SPARQL Protocol and RDF Query Language ). RDF has a standardized process of data definition; in other words, data define as subject-predicate-object triples. OWL is a continuation of RDF in which ontologies can be defined that describe complicated relationships between concepts in a domain. SPARQL query language is a powerful tool which allows a user to read and process the data which is stored in RDF.

### **1.3 Role of Ontologies in Healthcare**

Semantic Web is based on ontologies because ontologies are applied to provide the organized representation of knowledge within a specific sphere. An ontology defines classes, relationship and properties that define real life concepts. With ontologies, information can be exchanged and interpreted in a similar way by other systems and this improves data interoperability and integration. The effective management of this information is essential to improve healthcare services and ensure that the work with the patients is effective.

A project was developed using the Protégé ontology editor to develop a healthcare ontology. The ontology has represented key concepts in the health sector including patients, physicians, diseases, medicines, hospitals and appointments. To define the relationship between these objects object properties were implemented and to define the attributes of patient age and names data properties were introduced .

## **2. Domain Description**

### **2.1 Healthcare information systems**

Healthcare field is relevant in organisation and regulation of medical information about patients, medical staff, disease, treatment and medical facilities. A current healthcare system produces massive data in the form of healthcare providers, clinics, labs, and hospitals. These details will include patient history, diagnosis and treatment, prescription and clinic visit. Such information must be controlled effectively in discovery of effective health care services. There are several healthcare bodies that are interlinked with each other.

### **2.2 Key healthcare entities**

Patients belong to the cohort of people who seek the healthcare services due to their disease or health problems. Doctors in the Medical field are individuals in a healthcare business whose duties involve diagnosing, administering of drugs and treatment. Hospitals are the health care provision facilities. Appointments are pre-arranged relationships between the patients and medical workers. The term treatment refers to medical procedures or interventions that are designed to improve the health of the patients.

### **2.3 Relationships between healthcare entities**

The inclusion of these in this ontology was in the form of a type of care domain. The relationships between such classes were defined by using object properties. An example to this is when a patient has been diagnosed with an illness, the doctor he meets prescribes a medication to him or when he has a date with a doctor in a hospital. This renders querying and retrieving of knowledge in SPARQL and smart.

As an illustration, one will be in a position to acquire the details about patients that have been diagnosed to have some specific diseases, to determine who prescribes a specific medication to patients or the doctors who are treating patients in some hospitals. That proves the Semantic Web technologies can make the process of managing information and discovering knowledge possible in healthcare institutions.

## **3. Domain Model**

The healthcare ontology developed in this project can be considered the most important objects and relations that occur in a healthcare information system. The ontology is drawn on the

foundation of Protégé ontology editor and this is carried out with the help of Web ontology language (OWL). An entity model is such that the healthcare knowledge is in structured and machine readable form so as to easily query and analyze the information using Semantic web tools such as SPARQL. Ontology contains important healthcare concepts including patients, physicians, diseases, medicines, hospitals, and appointments.

The patients are able to pay a visit to hospitals, visit the physicians, and will be treated or prescribed medication. The type of reflection is the Doctor one, and the professional provides with the diagnosis of a disease or the treatment or the prescription of medicine. The ontology too defines Patient and Doctor as subclass of Person and hence will consider a logical hierarchy between the entities which will be a similarity of the real world relationship between the two entities. There are also other classes like Hospital, Disease, Medicine, Appointment, Treatment and MedicalRecord that are critical classes that have been included in the ontology.

The type of healthcare institutions where patients receive medical services is known as the Hospital class (Sajith Ranatunga et al., 2025). The Disease type is a medical disorder, which affects the patients and needs treating or diagnosing. Medicine category is a representation of drug materials which are used to treat a disease or administer symptoms. The Appointment class is a course of action that is carried out amid the patients and the medical professionals. The Treatment class involves medical practices or actions of improving on the health of patients. And finally, there is the MedicalRecord which is a historical information on patient of the medical record.

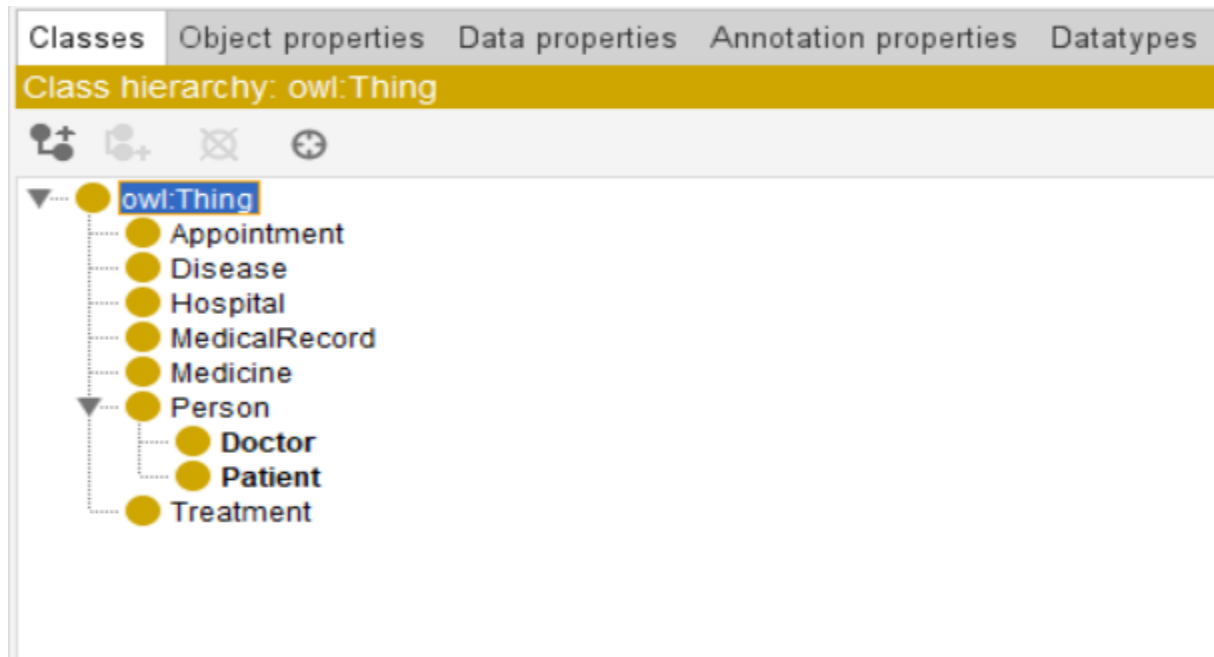


Figure 1: Healthcare ontology class hierarchy

Object properties were created to describe the association between the classes in the ontology. These properties are used to allow organizations within the healthcare system to interact with each other. An example is the `diagnosedWith` property which defines what exists between a `Patient` and a `Disease` in the sense that a patient has been diagnosed to have a particular medical condition. The association of a `Doctor` to a `Medicine`, which is the medicine prescribed by a doctor, is a property, `prescribedMedicine`.

`HasAppointment` property refers to a patient and an `Appointment` indicating that a patient has a definite appointment. An `appointmentAt` property is a reference between an `Appointment` and a `Hospital` which shows the location of the appointment. They help in simulation of the existing real world connections among medical objects.

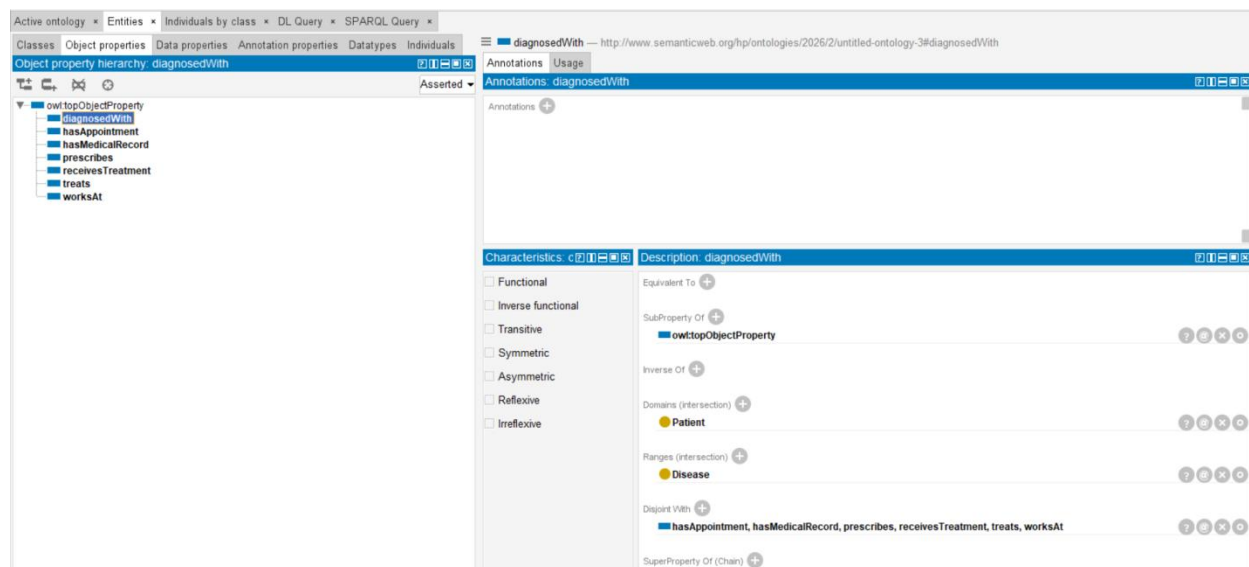


Figure 2: Object properties of diagnosed with

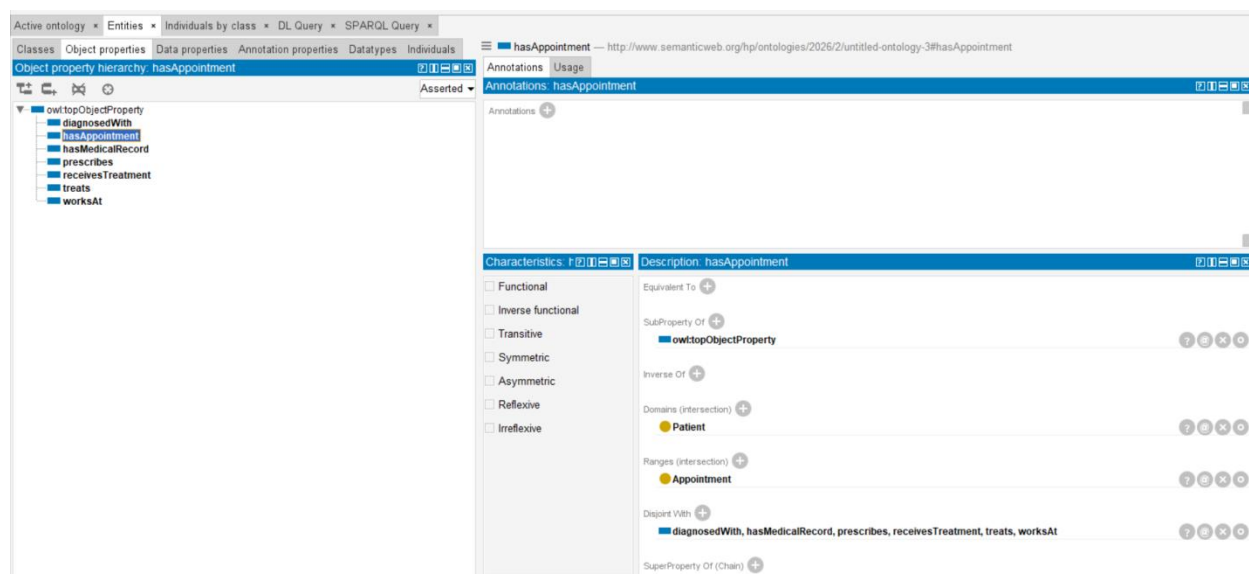


Figure 3: Object properties of hasAppointment

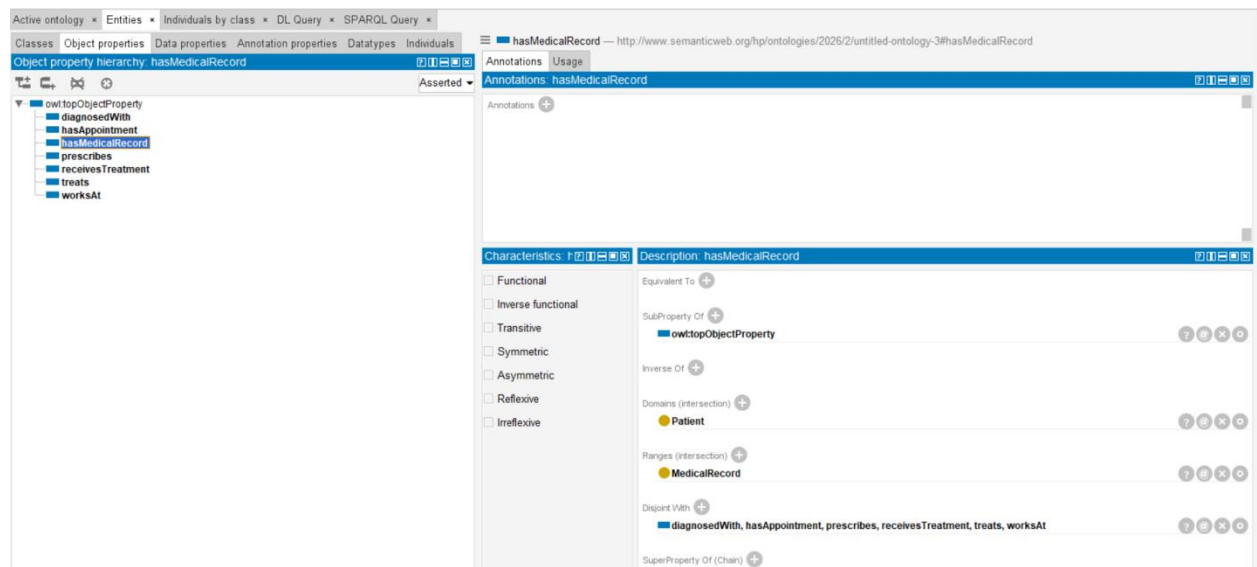


Figure 4: Object properties of hasMedicalRecord

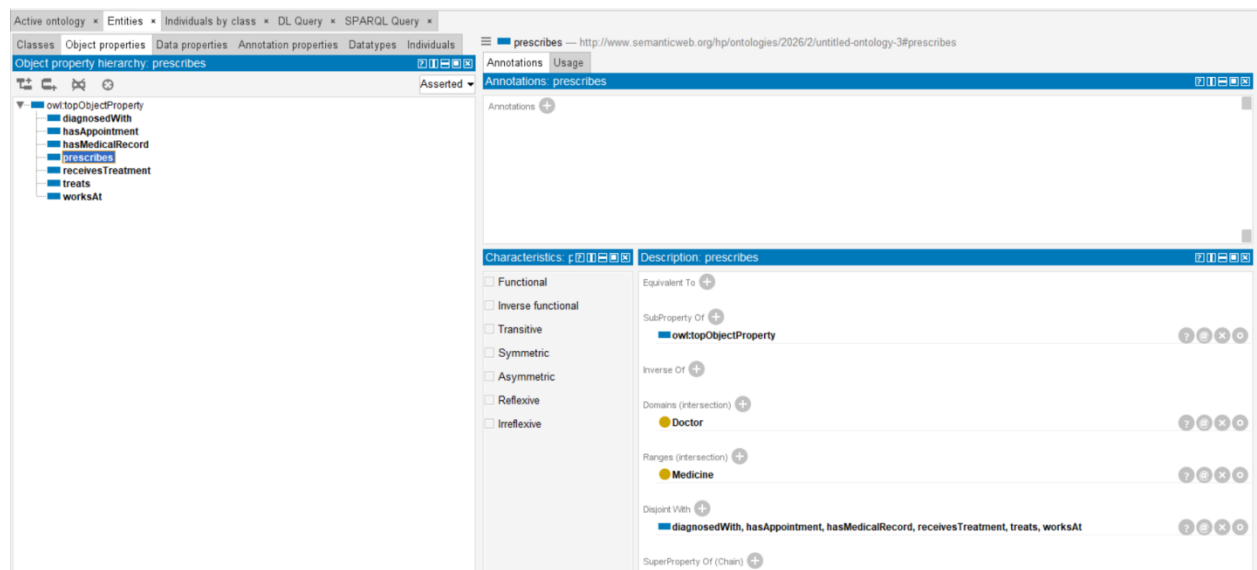


Figure 5: Object properties of prescribes

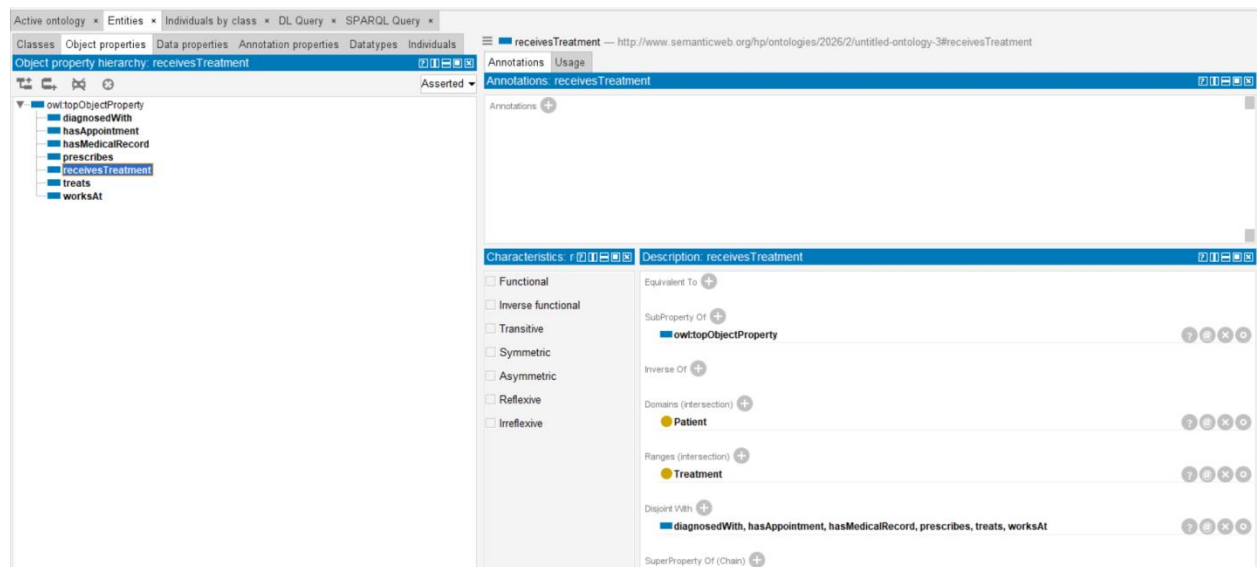


Figure 6: Object property of receives treatment

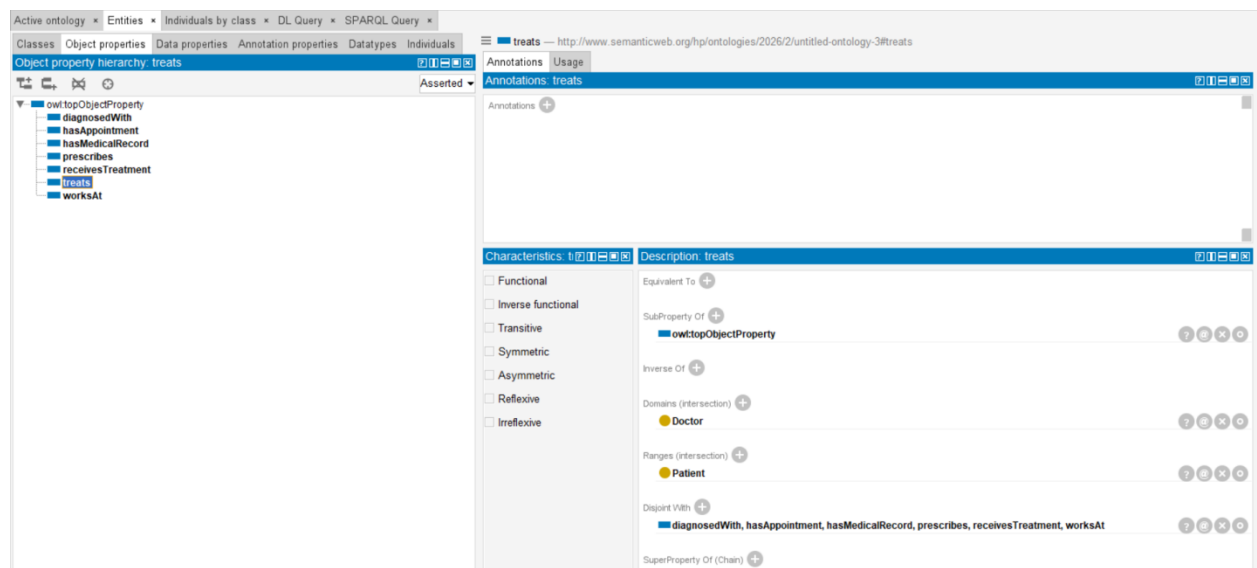


Figure 7: Object properties of treats

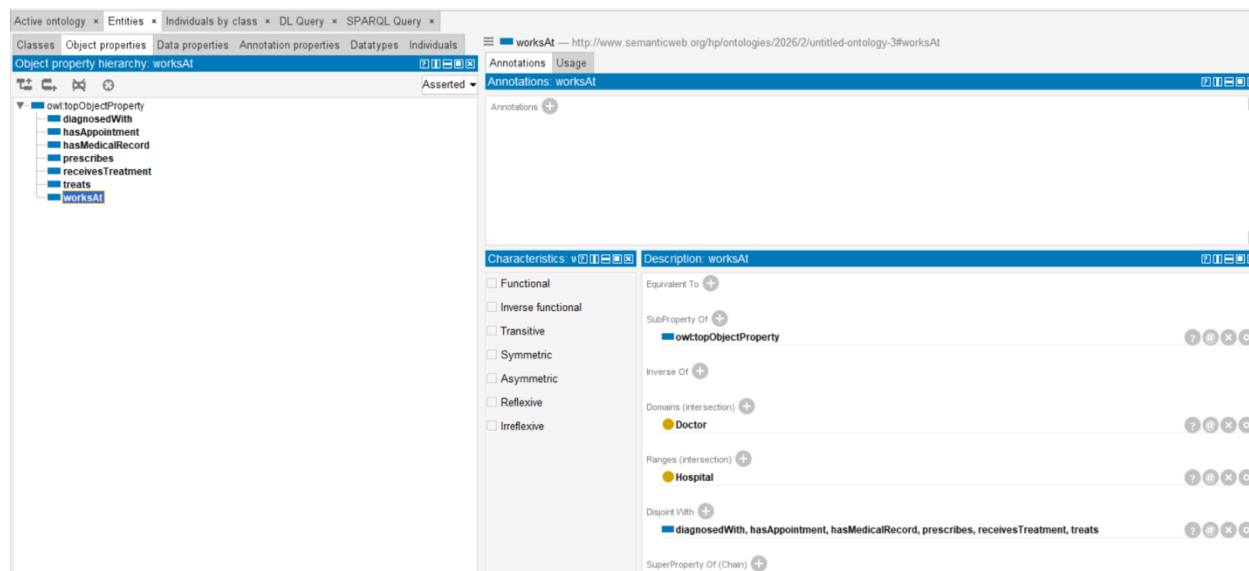


Figure 8: Object properties WorksAt

In addition to the properties of objects also, the ontology has data properties. Properties of data depict attributes of individuals. An example is a data property patientAge and its value is the age of a patient which can be attributed to an integer. Similarly, people can also be described using other forms of information that can be stored in the form of a text music in the properties such as medicineName or doctorName (Delgoshaei, Heidarinejad and Austin, 2022).

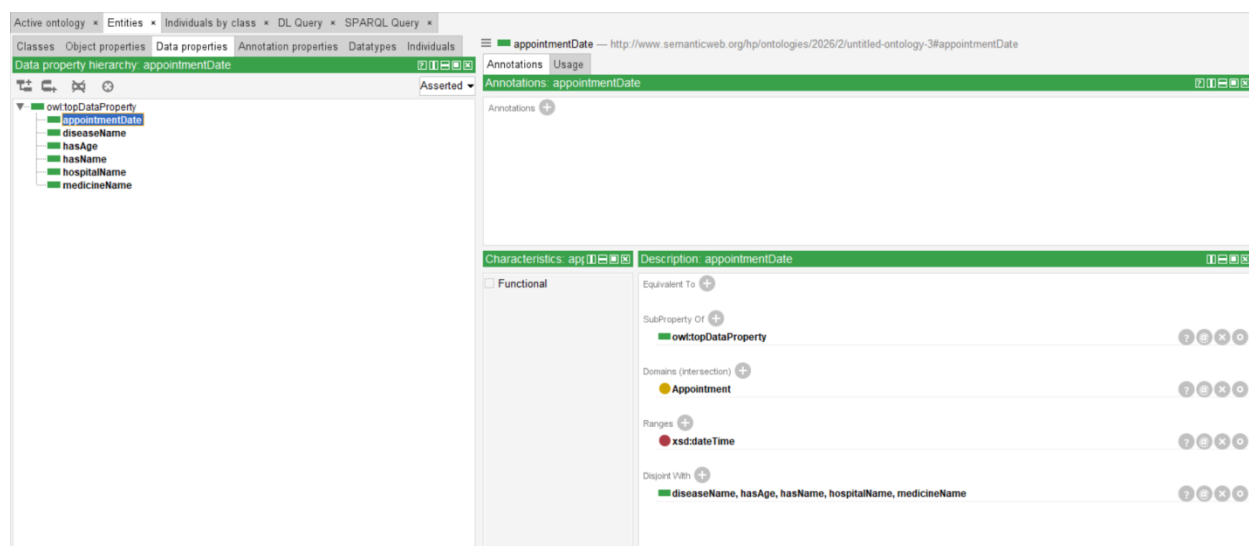


Figure 9: Data properties of appointmentDate

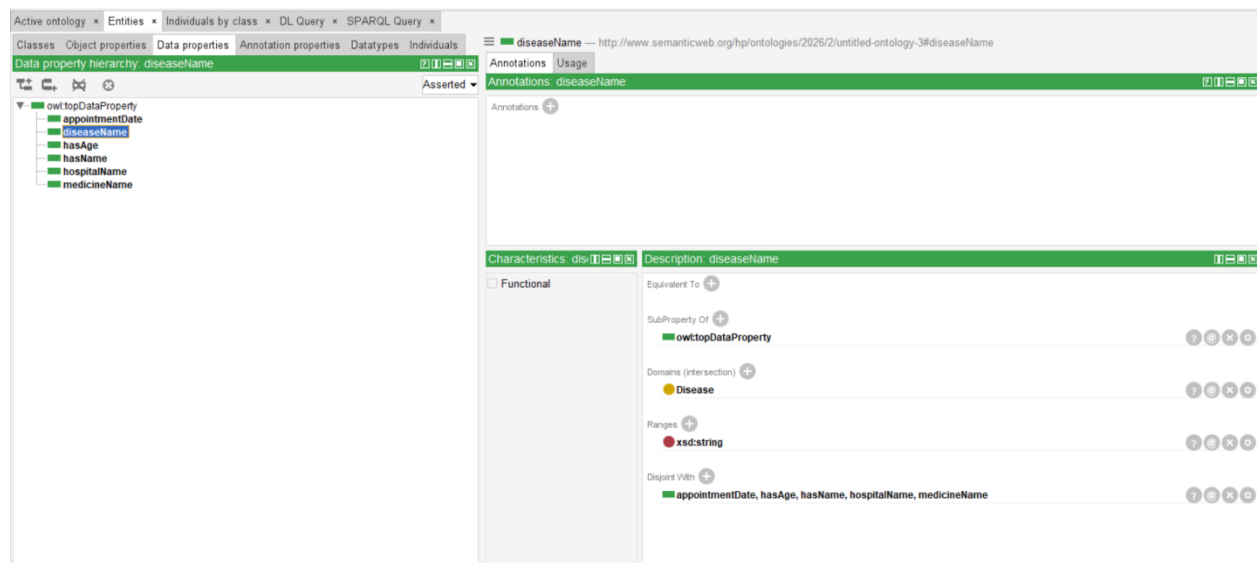


Figure 10: Data properties of disease name

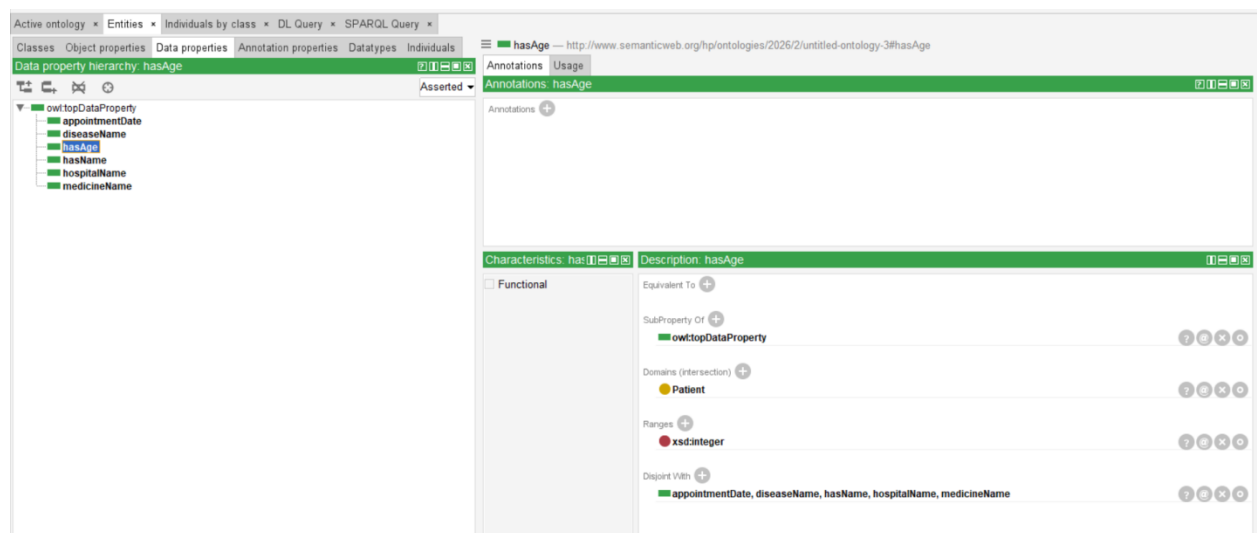


Figure 11: Data properties of hasAge

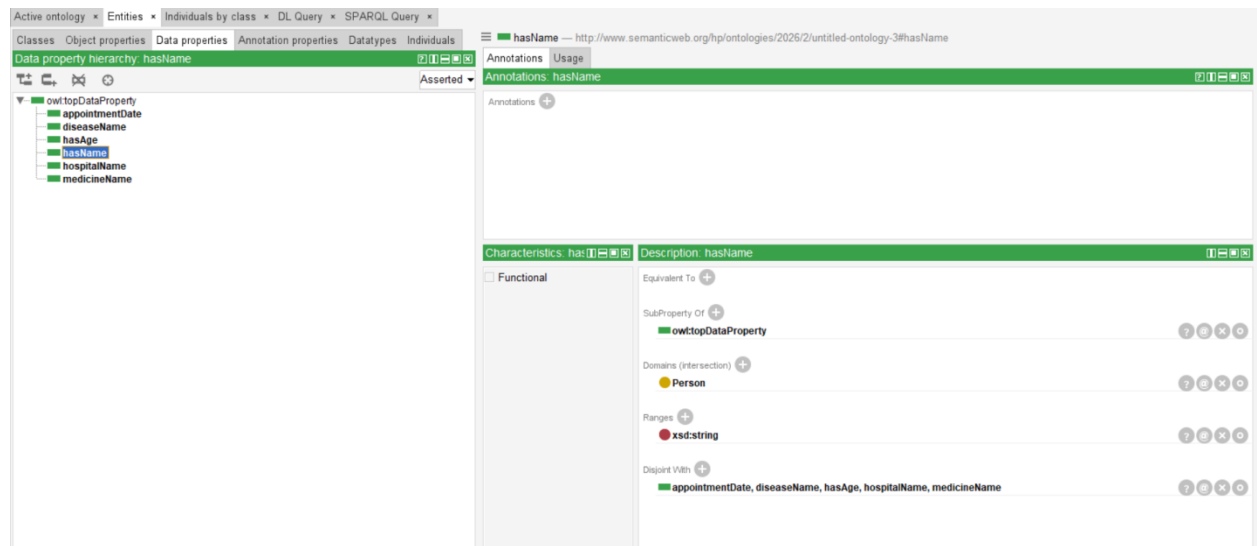


Figure 12: Data properties of hasName

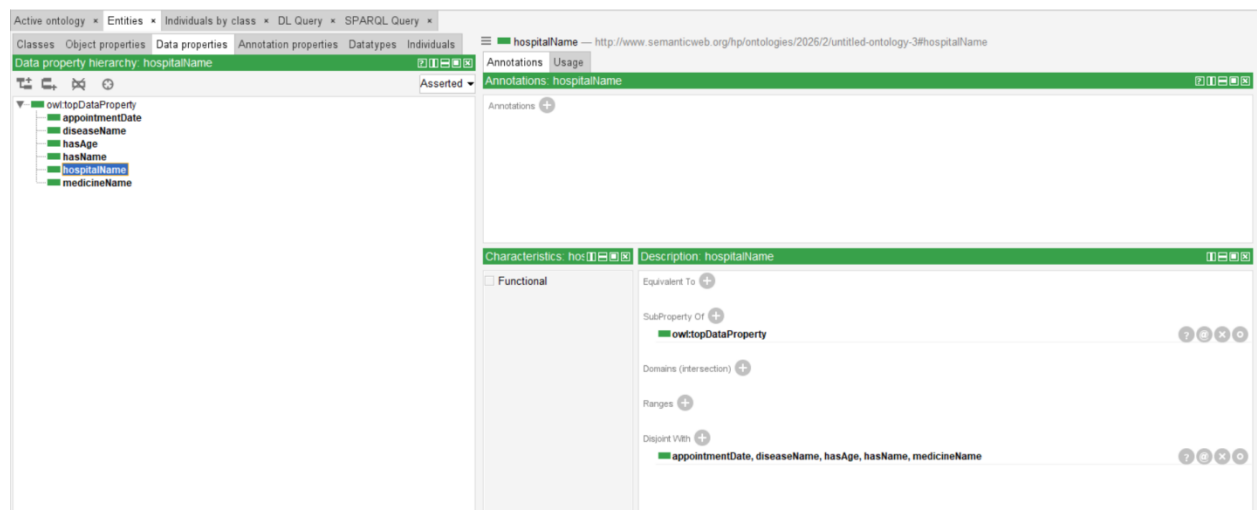


Figure 13: Data properties of hospitalName

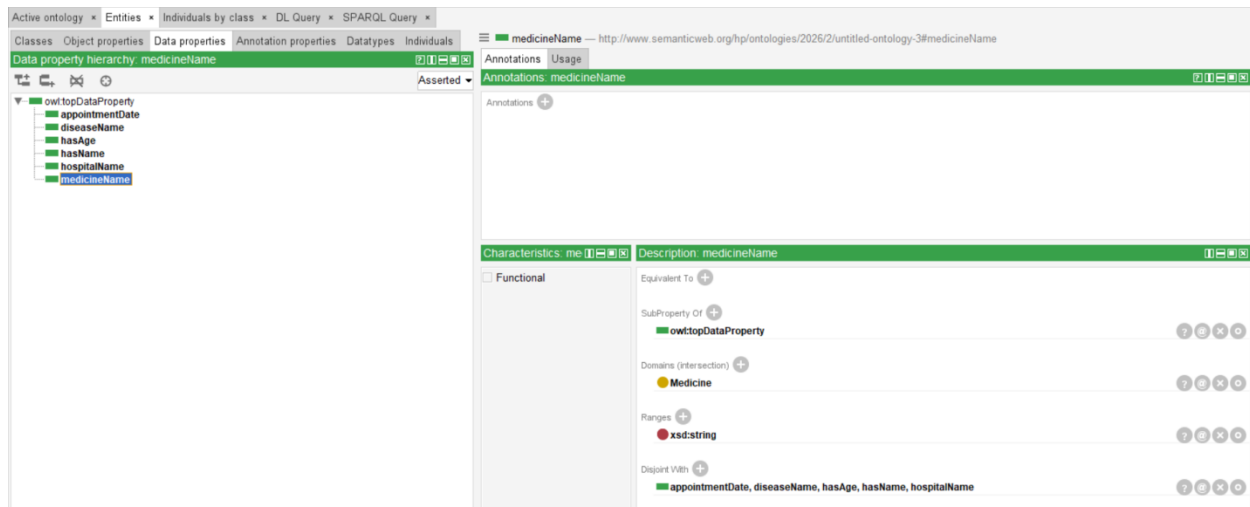


Figure 14: Data properties of medicineName

Entities were constructions that were illustrations of some of the examples of classes shown in the ontology. A good example would be a single individual JohnPatient who is a patient to the healthcare system. The character's name DrSmith is an epitome of a doctor who is a service provider to the patients. Diabetes type is an ailment to the patient, and Insulin is a medication that is utilized to manage the disorder . Other characters such as CityHospital and Appointment1 were created to describe the healthcare institutions and medical appointments.

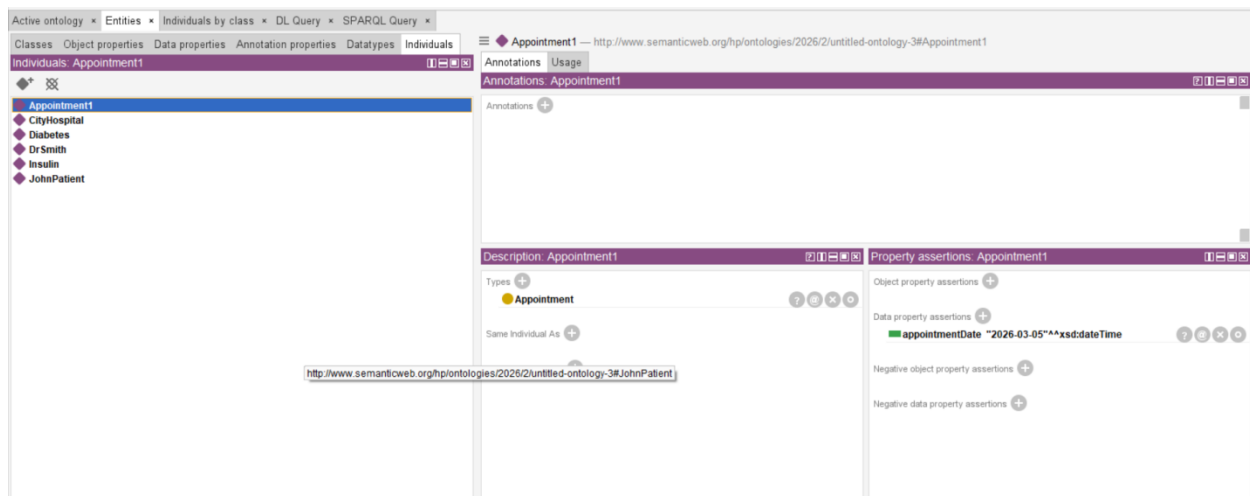


Figure 15: Individuals of Appointment

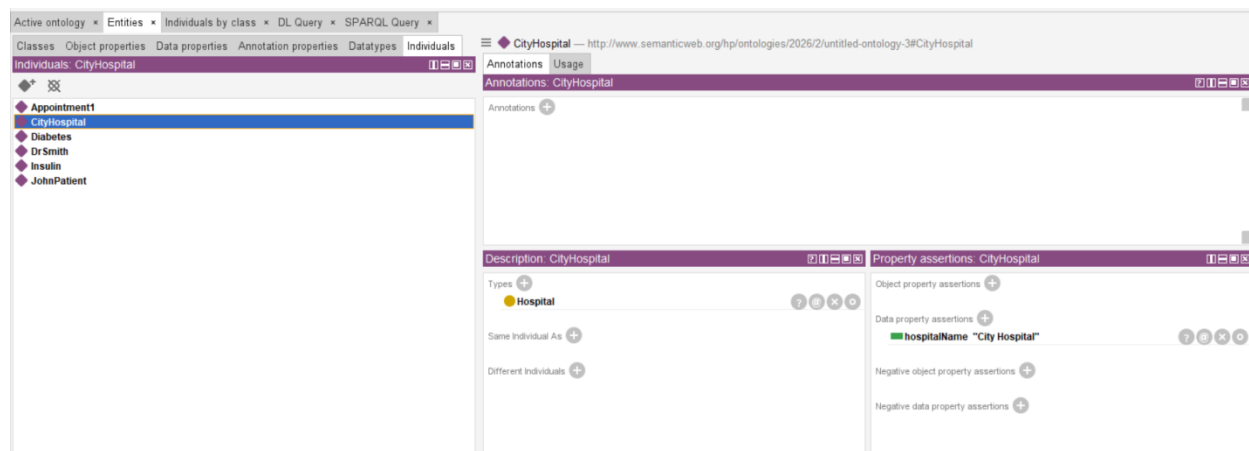


Figure 16: Individuals of CityHospital

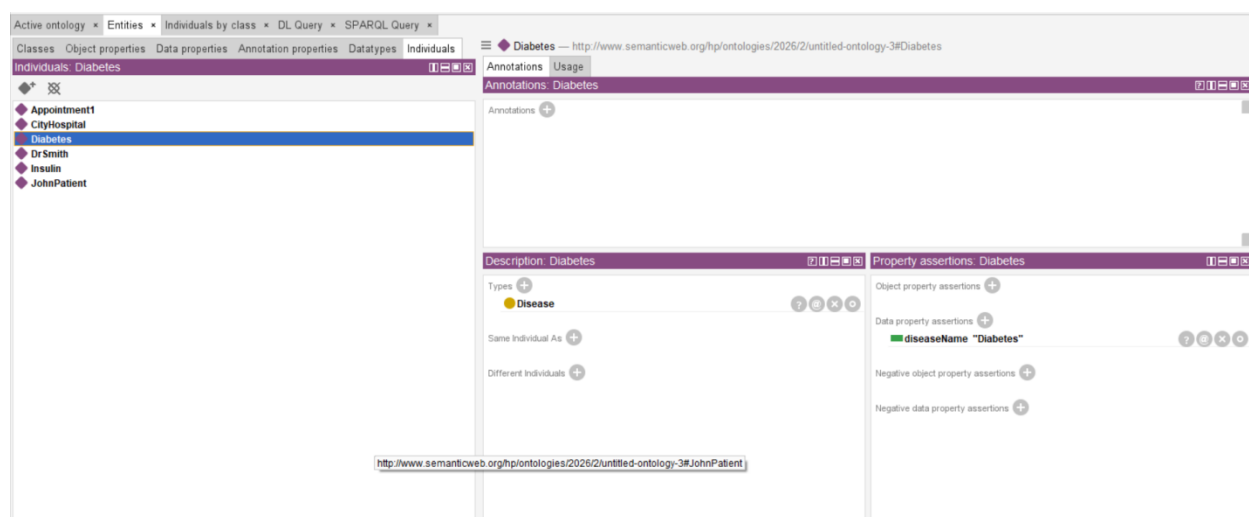


Figure 17: Individuals of Diabetes

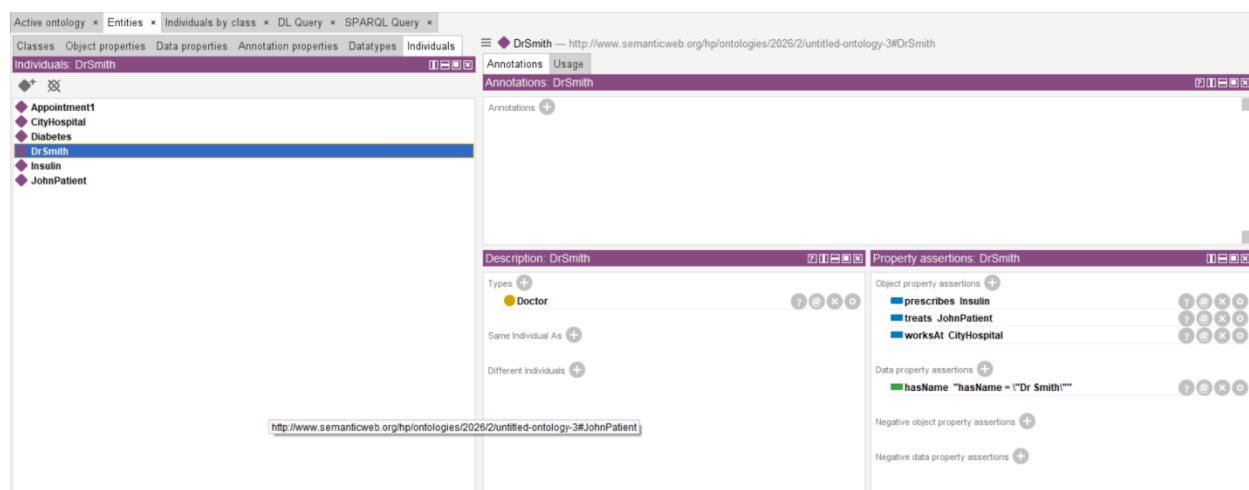


Figure 18: Individuals of DrSmith

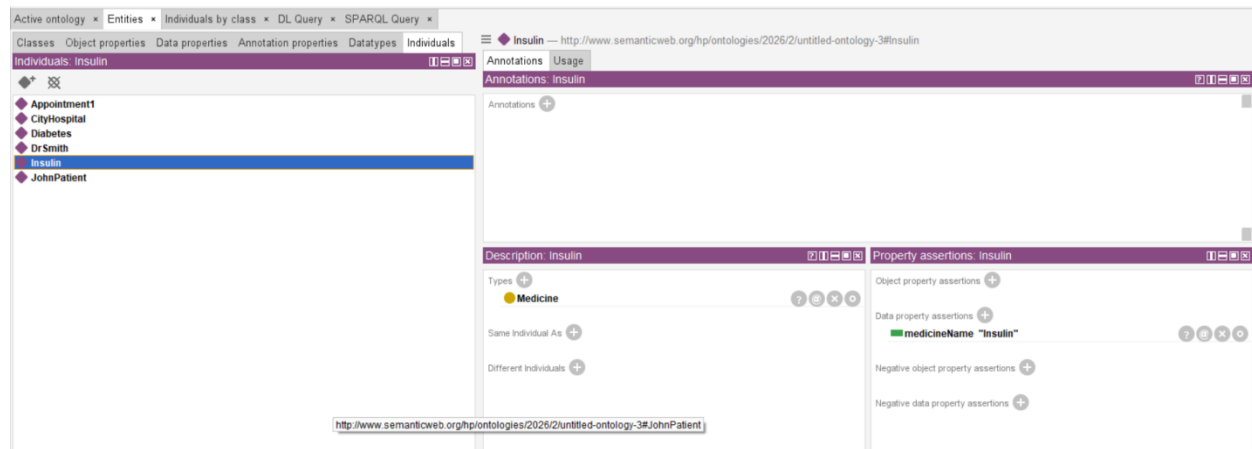


Figure 19: Individuals of Insulin

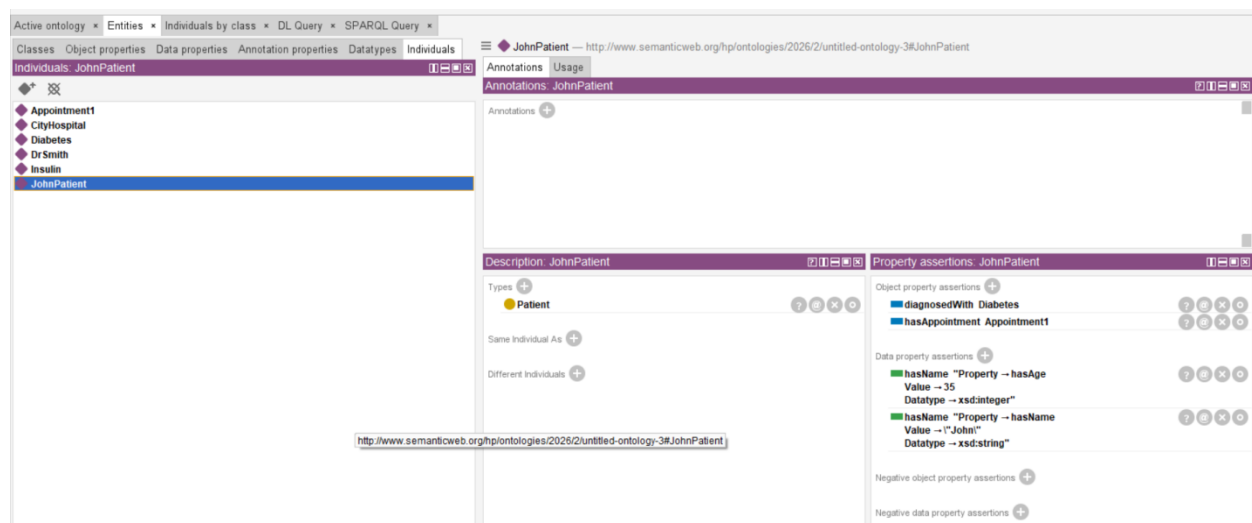


Figure 20: Individuals in JohnPatient

Object property assertions were the next ones to determine the relations between individuals. An example, the patient whose name is JohnPatient is attached to the disease Diabetes, through Diabetes property, which states that the patient has been detected with diabetes. Similarly, the Insulin drug is linked to the doctor, DrSmith using the prescribedMedicine property.

Protégé reasoner was used to check the structure of ontology as to whether it is logically valid. The ontology is verified by the reasoner both as to whether it contains inconsistencies or otherwise, and it also establishes whether or not people have been correctly classified in terms of adhering to

the respective relationships described. The real execution of the reasoner ensured the consistency of construct of ontology, as well as proper specification of all the relationships between healthcare entities.

The figure displays three sequential screenshots of the Protégé interface, each showing a DL query being executed. Each screenshot has a yellow header bar labeled "DL query:". Below this, there is a section titled "Query (class expression)" with a text input field. Underneath the input field are two buttons: "Execute" and "Add to ontology". The bottom section of each screenshot is titled "Query results" and shows the output of the query.

**First Screenshot:** The query input is "Patient". The "Query results" section shows "Instances (1 of 1)" with a single result, "JohnPatient", displayed in a blue bar with a red diamond icon and a help icon (?) on the right.

**Second Screenshot:** The query input is "Doctor". The "Query results" section shows "Instances (1 of 1)" with a single result, "DrSmith", displayed in a light blue bar with a red diamond icon and a help icon (?) on the right.

**Third Screenshot:** The query input is "Disease". The "Query results" section shows "Instances (1 of 1)" with a single result, "Diabetes", displayed in a light blue bar with a red diamond icon and a help icon (?) on the right.

Figure 21: Reasoner execution in Protégé

Generally, the healthcare domain model is able to represent important healthcare objects and their connection. Using the organization of the knowledge in the field of healthcare in the OWL, the ontology is the foundation of semantic querying and the discovery of knowledge through the use of the SPARQL.

#### 4. SPARQL queries

The standard query language that is applied in information retrieval and manipulation of information stored in RDF and OWL ontologies is known as SPARQL (SPARQL Protocol and RDF Query Language). It enables the user to inquire a semantic data through defining patterns appropriate to correspond to relationships among entities in the ontology.

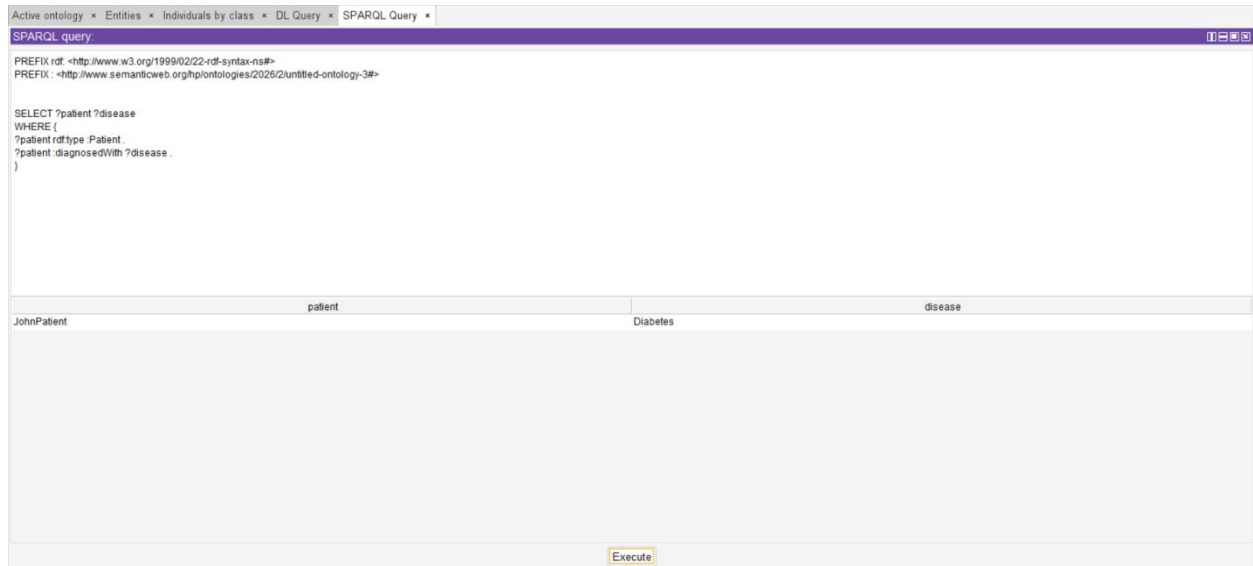


Figure 22: Patient disease relationship query

The initial query obtains all the patients and the illness that they are diagnosed with. This query contains two triple patterns in the WHERE clause, thus meeting the criteria of a typical SPARQL query that contains several conditions. The output indicates that, the patient JohnPatient has been diagnosed with the disease Diabetes and this indicates how relationships between healthcare organizations can be retrieved .

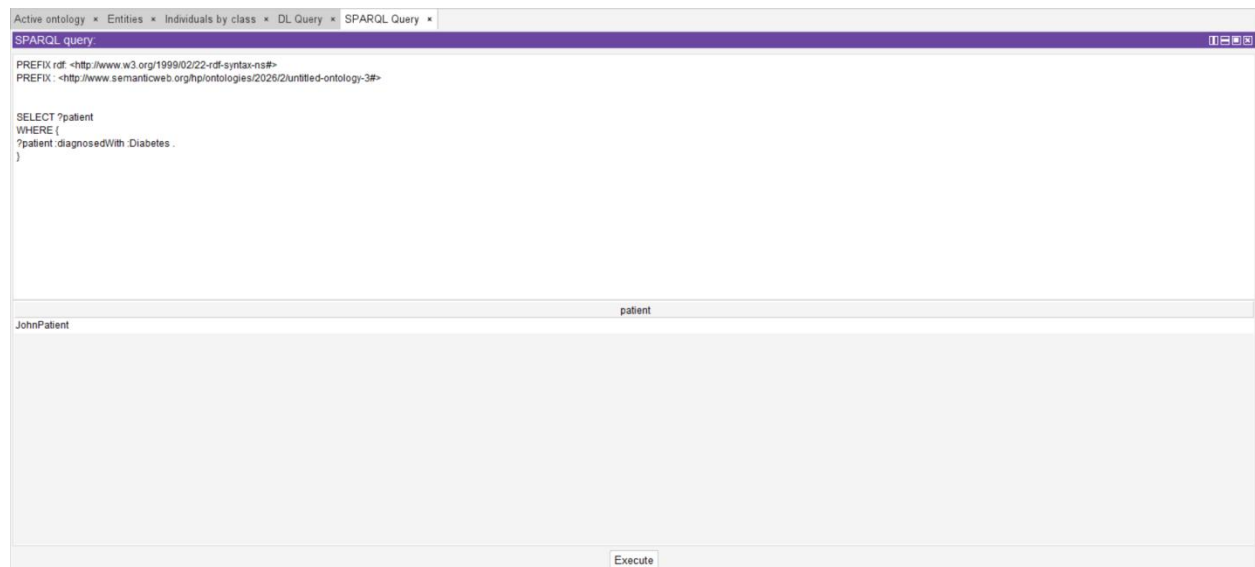


Figure 23: SPARQL Query Retrieving Patient Diagnosed with Diabetes

In this case Diabetes the second query obtains possession of the patient of a particular illness. The system will provide the outcome of a patient with the disease directly mentioned in query.

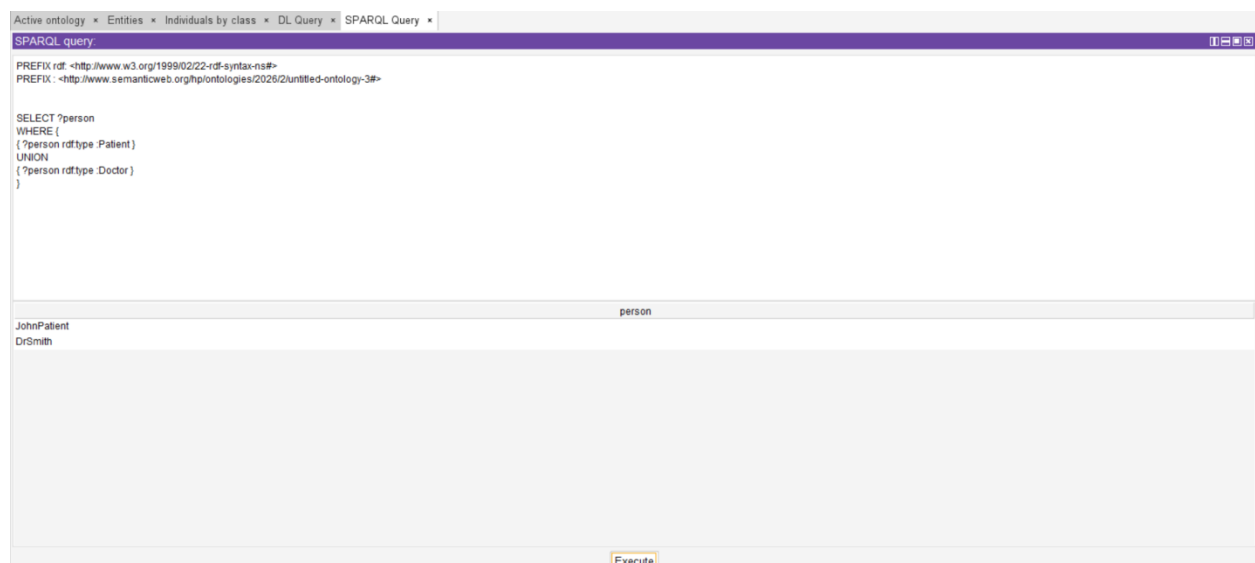


Figure 24: Union Query to find Patients and Doctors

The usage of UNION operator is represented in the third query. This query will give people of the type Patient or the Doctor. It is going to lead to the demonstration of both JohnPatient and DrSmith that indicates that UNION can combine the output of the diverse classes.

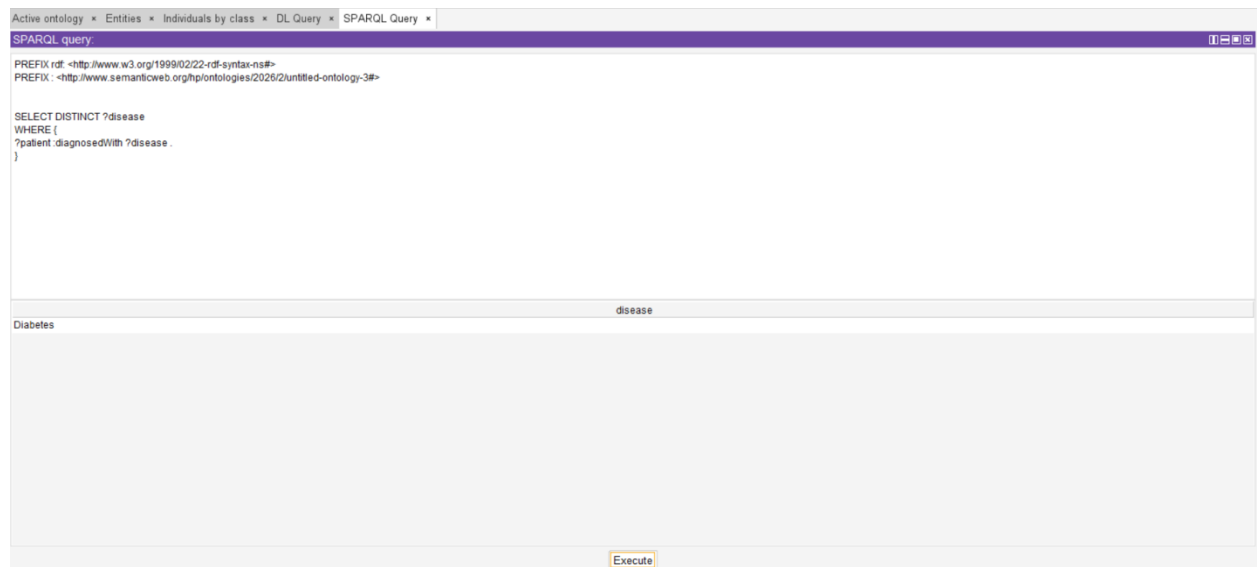


Figure 25: Query with the use of DISTINCT to access distinct diseases

The fourth query uses the DISTINCT keyword to retrieve unique diseases in the ontology in order to eliminate the similar values in the response.

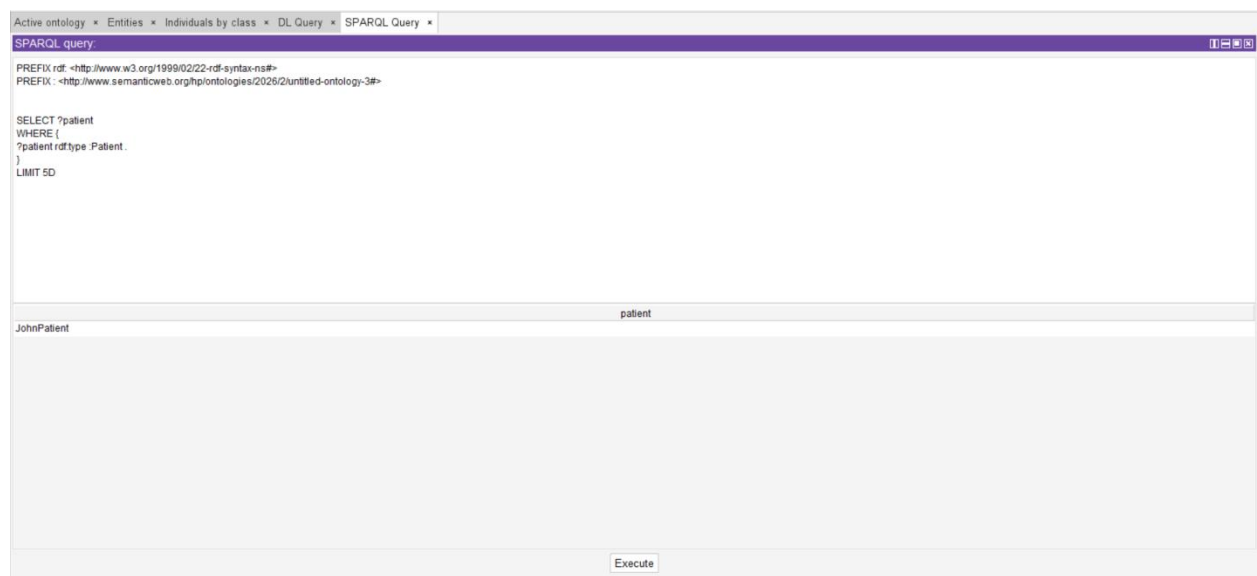


Figure 26: Query with LIMIT Clause

The fifth query demonstrates the use of THE LIMIT clause which is used to enforce limit on the number of the results that the query is going to fetch. It is handy where one has to deal with heavy data.

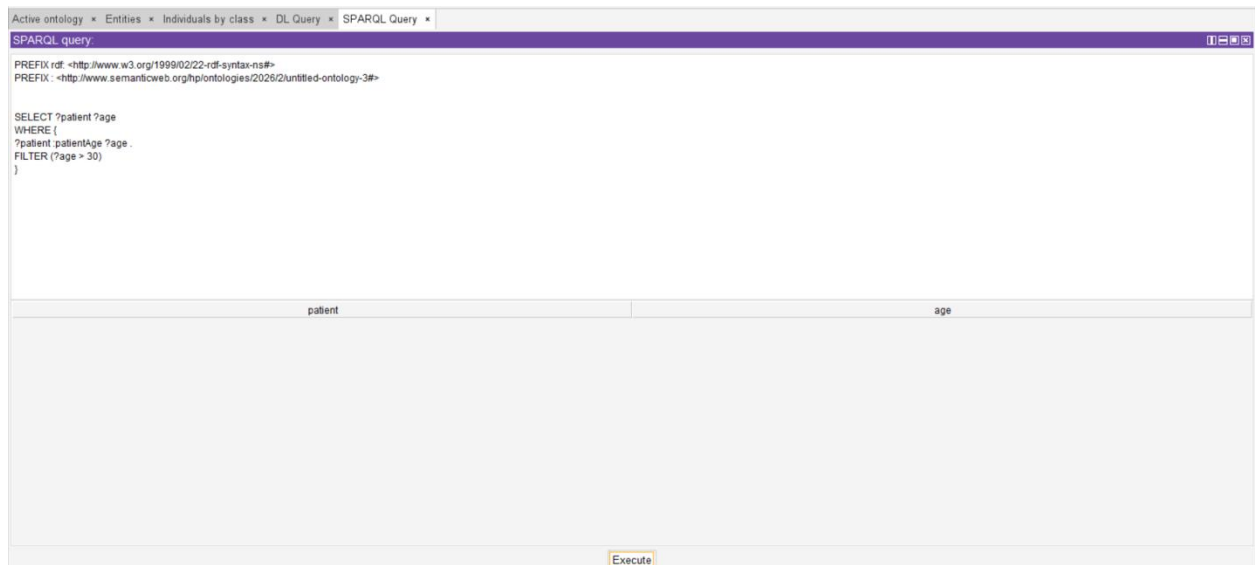


Figure 27: Query Sharing the same FILTER Condition

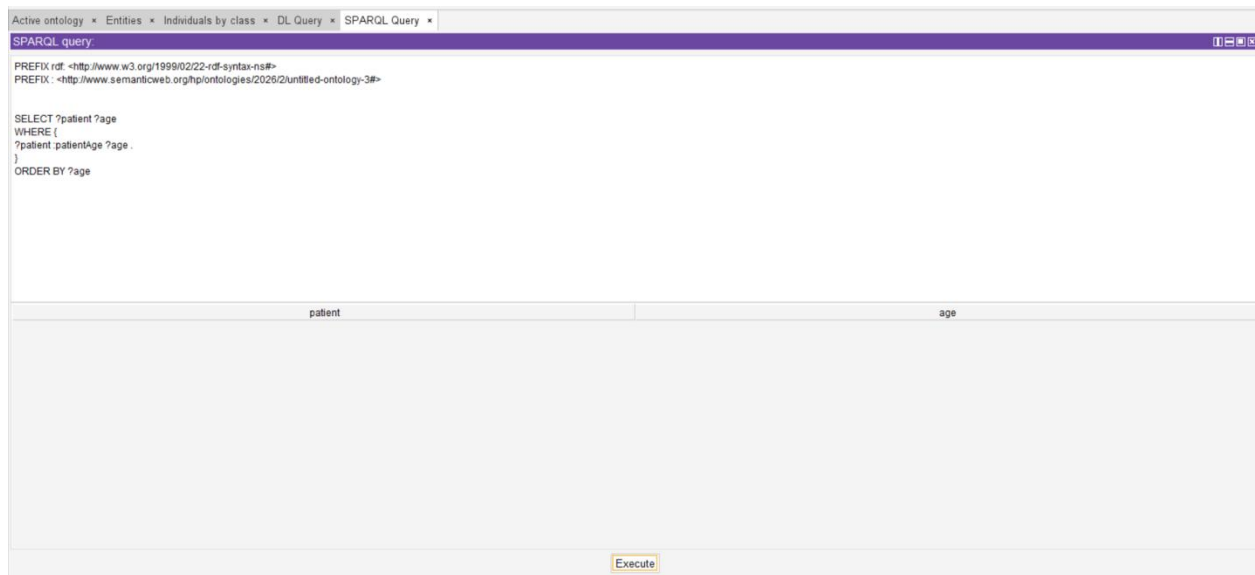


Figure 28: ORDER BY Query to Sort Query Results

The sixth query relies on FILTER condition to identify the patients whose age meets a specified condition. The seventh query uses ORDER BY to filter the messages of patients according to their age.

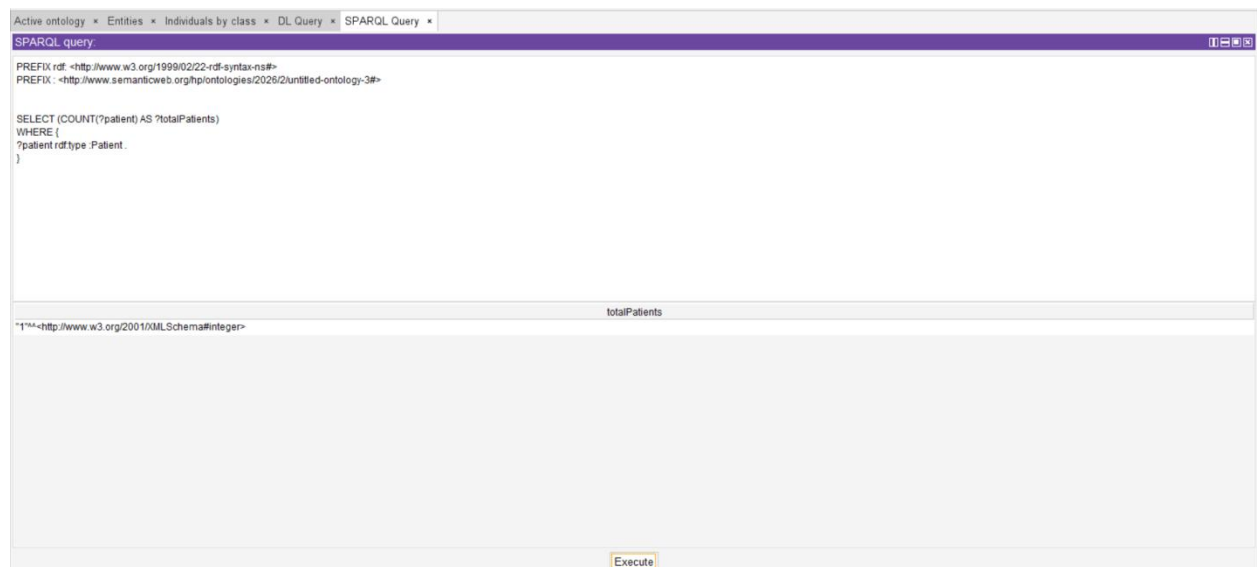


Figure 29: Count query

The other questions are querying links between the physicians and pharmaceuticals and the visits of patients and health centres. Finally, an aggregate query that is used to compute the aggregate total number of patients in the ontology; the query contains the COUNT keyword.

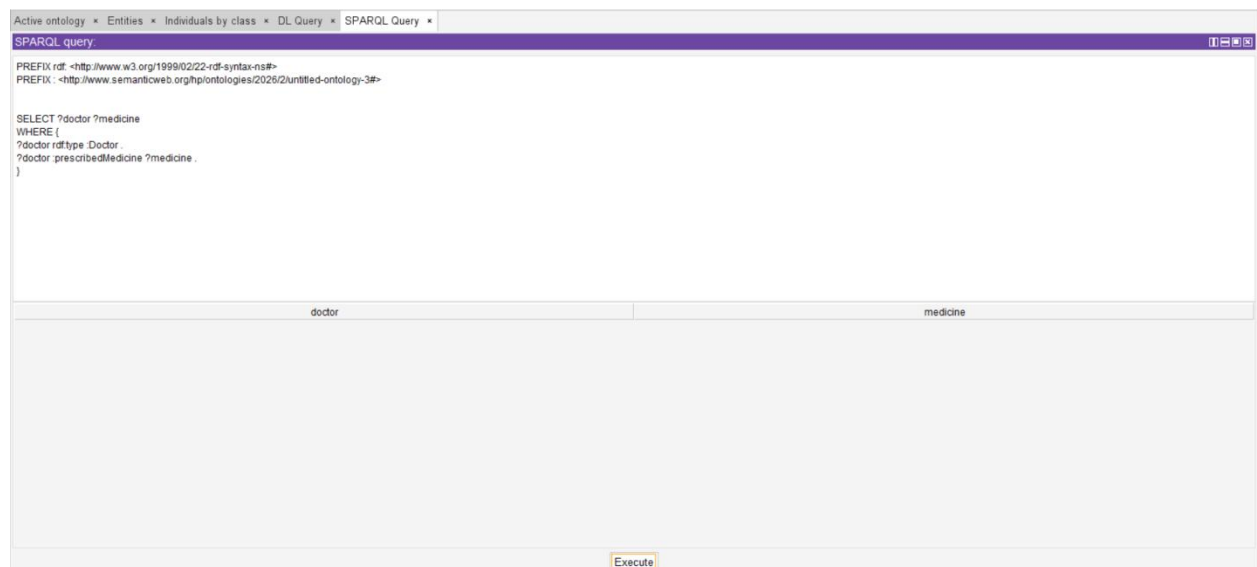


Figure 30: Query with doctor medicine

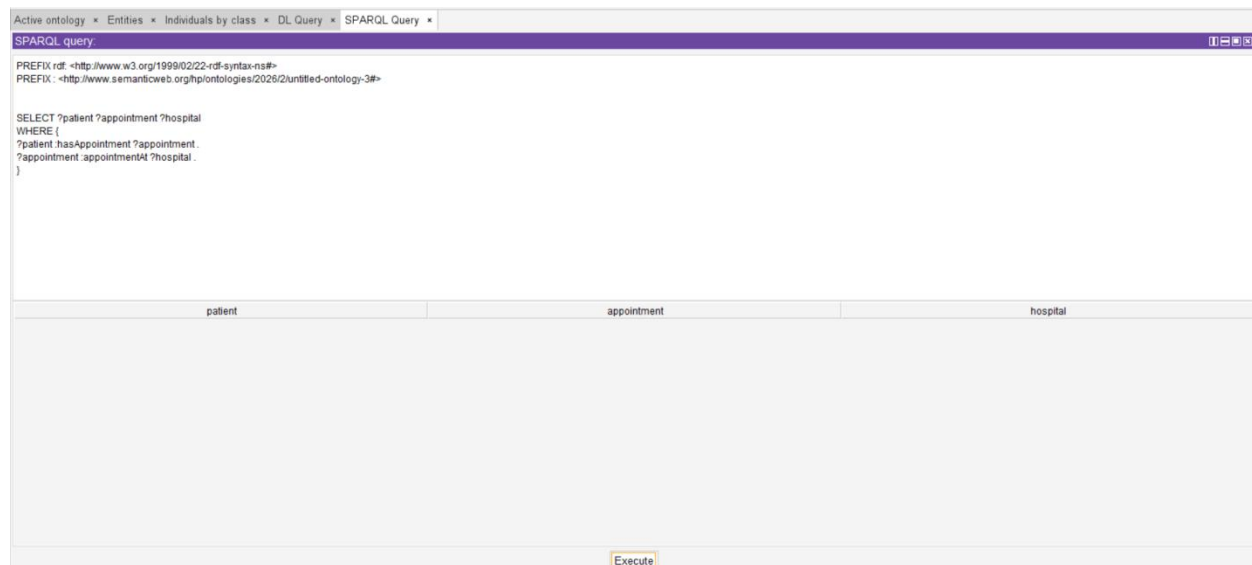


Figure 31: Query with patient appointment

These questions are indicative in the ability of the SPARQL to retrieve and analyze semantic healthcare data efficiently.

## 5. Conclusion

The project is directed to development of a healthcare ontology according to Semantic Web tools such as OWL and SPARQL. An ontology which models all the important healthcare entities like patients, physicians, illnesses, medications, hospitals and medical appointments was created using protégé ontology editor. The relationship between these entities was described using object properties as compared to data properties which represented age of patients and names of medicine.. These results indicate that Semantic Web technologies can be utilized in the field of effective knowledge presentation, information search and exchange in healthcare information systems.

Even though the ontology successfully got implemented, there are some limitations. The ontology is merely a case of healthcare with few classes and individuals that is simplified. The real health systems include more intricate relationships of patient medical history, laboratory results and other systems interaction (hospital multi-hospital). Subsequent application might further expand the ontology with the incorporation of standard healthcare ontologies, like SNOMED CT or HL7 and

larger data sets. This would enhance scalability, interoperability and applicability of the healthcare knowledge model in the real world.

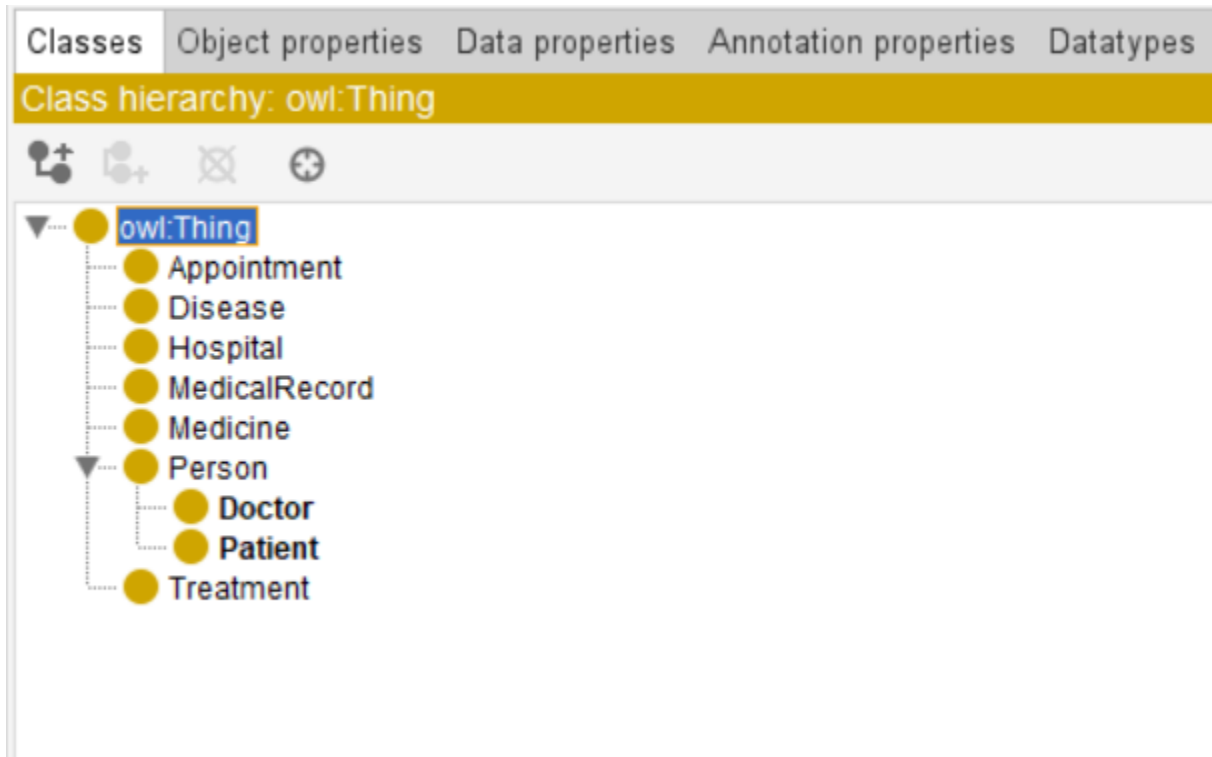
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## Appendix



occurs. They assist in replicating the existing real world linkages among healthcare objects.

The screenshot shows a web-based ontology editor interface. The top navigation bar includes 'Active ontology', 'Entities', 'Individuals by class', 'DL Query', and 'SPARQL Query'. The 'Object properties' tab is active, and the title bar reads 'Object property hierarchy: diagnosedWith'. The left sidebar shows a tree view of the object property hierarchy, with 'diagnosedWith' selected. The main area displays the 'diagnosedWith' property. The 'Annotations' section is empty. The 'Characteristics' section shows a list of checkboxes for property characteristics: 'Functional', 'Inverse functional', 'Transitive', 'Symmetric', 'Asymmetric', 'Reflexive', and 'Irreflexive'. The 'Description' section shows the 'diagnosedWith' property with its domain and range. The domain is 'Patient' and the range is 'Disease'. The 'Disjoint With' section shows a list of properties: 'hasAppointment', 'hasMedicalRecord', 'prescribes', 'receivesTreatment', 'treats', and 'worksAt'.

Active ontology: Entities: Individuals by class: DL Query: SPARQL Query

Classes: Object properties: Data properties: Annotation properties: Datatypes: Individuals

Object property hierarchy: **hasAppointment**

Annotations: **hasAppointment**

Annotations: **hasAppointment**

Characteristics: **hasAppointment**

Functional: ☐ Inverse functional: ☐ Transitive: ☐ Symmetric: ☐ Asymmetric: ☐ Reflexive: ☐ Irreflexive: ☐

Description: **hasAppointment**

Equivalent To: **owl:topObjectProperty**

SubProperty Of: **owl:topObjectProperty**

Inverse Of: **owl:topObjectProperty**

Domains (Intersection): **Patient**

Ranges (Intersection): **Appointment**

Disjoint With: **diagnosedWith, hasMedicalRecord, prescribes, receivesTreatment, treats, worksAt**

SuperProperty Of (Chain): **owl:topObjectProperty**

Active ontology: Entities: Individuals by class: DL Query: SPARQL Query

Classes: Object properties: Data properties: Annotation properties: Datatypes: Individuals

Object property hierarchy: **hasMedicalRecord**

Annotations: **hasMedicalRecord**

Annotations: **hasMedicalRecord**

Characteristics: **hasMedicalRecord**

Functional: ☐ Inverse functional: ☐ Transitive: ☐ Symmetric: ☐ Asymmetric: ☐ Reflexive: ☐ Irreflexive: ☐

Description: **hasMedicalRecord**

Equivalent To: **owl:topObjectProperty**

SubProperty Of: **owl:topObjectProperty**

Inverse Of: **owl:topObjectProperty**

Domains (Intersection): **Patient**

Ranges (Intersection): **MedicalRecord**

Disjoint With: **diagnosedWith, hasAppointment, prescribes, receivesTreatment, treats, worksAt**

SuperProperty Of (Chain): **owl:topObjectProperty**

Active ontology: Entities: Individuals by class: DL Query: SPARQL Query: **prescribes** — http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#prescribes

Classes: Object properties: Data properties: Annotation properties: Datatypes: Individuals: **prescribes**

Object property hierarchy: **prescribes**

Annotations: **prescribes**

Annotations: **prescribes**

Characteristics: **prescribes**

Description: **prescribes**

Equivalent To: **owl:topObjectProperty**

SubProperty Of: **owl:topObjectProperty**

Inverse Of: **owl:topObjectProperty**

Domains (Intersection): **Doctor**

Ranges (Intersection): **Medicine**

Disjoint With: **diagnosedWith, hasAppointment, hasMedicalRecord, receivesTreatment, treats, worksAt**

SuperProperty Of (Chain): **owl:topObjectProperty**

Active ontology: Entities: Individuals by class: DL Query: SPARQL Query: **receivesTreatment** — http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#receivesTreatment

Classes: Object properties: Data properties: Annotation properties: Datatypes: Individuals: **receivesTreatment**

Object property hierarchy: **receivesTreatment**

Annotations: **receivesTreatment**

Annotations: **receivesTreatment**

Characteristics: **receivesTreatment**

Description: **receivesTreatment**

Equivalent To: **owl:topObjectProperty**

SubProperty Of: **owl:topObjectProperty**

Inverse Of: **owl:topObjectProperty**

Domains (Intersection): **Patient**

Ranges (Intersection): **Treatment**

Disjoint With: **diagnosedWith, hasAppointment, hasMedicalRecord, prescribes, treats, worksAt**

SuperProperty Of (Chain): **owl:topObjectProperty**





Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

Classes ▾ Object properties ▾ Data properties ▾ Annotation properties ▾ Datatypes ▾ Individuals ▾

Data property hierarchy: hasName [?] [1] [2] [3] [4] [5]

Annotations: hasName [?] [1] [2] [3] [4] [5]

Annotations: hasName [?] [1] [2] [3] [4] [5]

Annotations: +

Characteristics: has [?] [1] [2] [3] [4] [5]

Functional

Description: hasName [?] [1] [2] [3] [4] [5]

Equivalent To +

SubProperty Of +

Domains (intersection) +

Ranges +

Disjoint With +

owl:topDataProperty

appointmentDate

diseaseName

hasAge

hasName

hospitalName

medicineName

Person

xsd:string

appointmentDate, diseaseName, hasAge, hospitalName, medicineName

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

Classes ▾ Object properties ▾ Data properties ▾ Annotation properties ▾ Datatypes ▾ Individuals ▾

Data property hierarchy: hospitalName [?] [1] [2] [3] [4] [5]

Annotations: hospitalName [?] [1] [2] [3] [4] [5]

Annotations: hospitalName [?] [1] [2] [3] [4] [5]

Annotations: +

Characteristics: hospitalName [?] [1] [2] [3] [4] [5]

Functional

Description: hospitalName [?] [1] [2] [3] [4] [5]

Equivalent To +

SubProperty Of +

Domains (intersection) +

Ranges +

Disjoint With +

owl:topDataProperty

appointmentDate

diseaseName

hasAge

hasName

hospitalName

medicineName

appointmentDate, diseaseName, hasAge, hasName, medicineName

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

Classes ▾ Object properties ▾ Data properties ▾ Annotation properties ▾ Datatypes ▾ Individuals ▾

Data property hierarchy: medicineName [?] [1] [2] [3] [4] [5]

Annotations: medicineName [?] [1] [2] [3] [4] [5]

Annotations: medicineName [?] [1] [2] [3] [4] [5]

Annotations: +

Characteristics: medicineName [?] [1] [2] [3] [4] [5]

Functional

Description: medicineName [?] [1] [2] [3] [4] [5]

Equivalent To +

SubProperty Of +

Domains (intersection) +

Ranges +

Disjoint With +

owl:topDataProperty

appointmentDate

diseaseName

hasAge

hasName

hospitalName

medicineName

Medicine

xsd:string

appointmentDate, diseaseName, hasAge, hasName, hospitalName

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Classes Object properties Data properties Annotation properties Datatypes Individuals

Individuals: Appointment1

Appointment1

CityHospital

Diabetes

DrSmith

Insulin

JohnPatient

Annotations Appointment1

Annotations

Description: Appointment1

Types

Appointment

Same Individual As

Property assertions: Appointment1

Object property assertions

Data property assertions

appointmentDate "2026-03-05"^^xsd:dateTime

Negative object property assertions

Negative data property assertions

<http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#JohnPatient>

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Classes Object properties Data properties Annotation properties Datatypes Individuals

Individuals: CityHospital

Appointment1

CityHospital

Diabetes

DrSmith

Insulin

JohnPatient

Annotations CityHospital

Annotations

Description: CityHospital

Types

Hospital

Same Individual As

Different Individuals

Property assertions: CityHospital

Object property assertions

Data property assertions

hospitalName "City Hospital"

Negative object property assertions

Negative data property assertions

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Classes Object properties Data properties Annotation properties Datatypes Individuals

Individuals: Diabetes

Appointment1

CityHospital

Diabetes

DrSmith

Insulin

JohnPatient

Annotations Diabetes

Annotations

Description: Diabetes

Types

Disease

Same Individual As

Different Individuals

Property assertions: Diabetes

Object property assertions

Data property assertions

diseaseName "Diabetes"

Negative object property assertions

Negative data property assertions

<http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#JohnPatient>

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Classes Object properties Data properties Annotation properties Datatypes Individuals

Individuals: DrSmith

- Appointment1
- CityHospital
- Diabetes
- Dr Smith
- Insulin
- JohnPatient

Annotations Usage

Annotations: DrSmith

Description: DrSmith

Types

- Doctor

Same Individual As

Different Individuals

Property assertions: DrSmith

Object property assertions

- prescribes Insulin
- treats JohnPatient
- worksAt CityHospital

Data property assertions

- hasName "hasName = 'Dr Smith'"

Negative object property assertions

Negative data property assertions

<http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#JohnPatient>

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Classes Object properties Data properties Annotation properties Datatypes Individuals

Individuals: Insulin

- Appointment1
- CityHospital
- Diabetes
- Dr Smith
- Insulin
- JohnPatient

Annotations Usage

Annotations: Insulin

Description: Insulin

Types

- Medicine

Same Individual As

Different Individuals

Property assertions: Insulin

Object property assertions

Data property assertions

- medicineName "Insulin"

Negative object property assertions

Negative data property assertions

<http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#JohnPatient>

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Classes Object properties Data properties Annotation properties Datatypes Individuals

Individuals: JohnPatient

- Appointment1
- CityHospital
- Diabetes
- Dr Smith
- Insulin
- JohnPatient

Annotations Usage

Annotations: JohnPatient

Description: JohnPatient

Types

- Patient

Same Individual As

Different Individuals

Property assertions: JohnPatient

Object property assertions

- diagnosedWith Diabetes
- hasAppointment Appointment1

Data property assertions

- hasName "Property -- hasAge Value -- 35 Datatype -- xsd:integer"
- hasName "Property -- hasName Value -- 'John' Datatype -- xsd:string"

Negative object property assertions

Negative data property assertions

<http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#JohnPatient>

DL query:

**Query (class expression)**

Patient

**Query results**

Instances (1 of 1)

|                                                                                               |                                                                                     |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  JohnPatient |  |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

DL query:

**Query (class expression)**

Doctor

**Query results**

Instances (1 of 1)

|                                                                                           |                                                                                     |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  DrSmith |  |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

DL query:

**Query (class expression)**

Disease

**Query results**

Instances (1 of 1)

|                                                                                              |                                                                                       |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  Diabetes |  |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query:   

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hpl/ontologies/2025/2/untitled-ontology-3#>

SELECT ?patient ?disease  
WHERE {  
 ?patient rdf:type :Patient .  
 ?patient :diagnosedWith ?disease .  
}

|             | patient | disease  |
|-------------|---------|----------|
| JohnPatient |         | Diabetes |

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query:   

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hpl/ontologies/2025/2/untitled-ontology-3#>

SELECT ?patient  
WHERE {  
 ?patient :diagnosedWith :Diabetes .  
}

|             | patient |
|-------------|---------|
| JohnPatient |         |

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#>  
  
SELECT ?person  
WHERE {  
 { ?person rdf:type :Patient }  
 UNION  
 { ?person rdf:type :Doctor }  
}

person

JohnPatient  
DrSmith

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#>  
  
SELECT DISTINCT ?disease  
WHERE {  
 ?patient :diagnosedWith ?disease .  
}

disease

Diabetes

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#>  
  
SELECT ?patient  
WHERE {  
 ?patient rdf:type Patient .  
}  
LIMIT 50

patient

JohnPatient

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#>  
  
SELECT ?patient ?age  
WHERE {  
 ?patient patientAge ?age .  
 FILTER (?age > 30)  
}

patientage

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#>

SELECT ?patient ?age  
WHERE {  
 ?patient :patientAge ?age .  
}  
ORDER BY ?age

| patient | age |
|---------|-----|
|---------|-----|

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hp/ontologies/2026/2/untitled-ontology-3#>

SELECT (COUNT(?patient) AS ?totalPatients)  
WHERE {  
 ?patient rdf:type :Patient .  
}

| totalPatients |
|---------------|
| 1             |

\*1^^http://www.w3.org/2001/XMLSchema#integer

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hpl/ontologies/2026/2/untitled-ontology-3#>  
  
SELECT ?doctor ?medicine  
WHERE {  
 ?doctor rdf:type :Doctor .  
 ?doctor prescribed ?medicine .  
}

| doctor | medicine |
|--------|----------|
|--------|----------|

Execute

Active ontology ▾ Entities ▾ Individuals by class ▾ DL Query ▾ SPARQL Query ▾

SPARQL query

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX : <http://www.semanticweb.org/hpl/ontologies/2026/2/untitled-ontology-3#>  
  
SELECT ?patient ?appointment ?hospital  
WHERE {  
 ?patient :hasAppointment ?appointment .  
 ?appointment :appointmentAt ?hospital .  
}

| patient | appointment | hospital |
|---------|-------------|----------|
|---------|-------------|----------|

Execute